

Appendices to Application 1 — FEC Methods

APPENDICES TO APPLICATION 1 — FEC METHODS

Filed as part of the provisional disclosure of APPLICATION-1-FEC-METHODS. Each appendix is incorporated in its entirety into the specification.

APPENDIX A — Prioritized Claim Set (Claims A–F, A4–A6) with Prior-Art Analysis

Patent Claims — Blockcast MoQ/MMTP/FEC Innovations

Prior Art Research Summary

Research conducted across RFC corpus, USPTO/EPO patent databases, academic literature, and ATSC/3GPP/DVB standards. Per-claim analysis below.

TOP 6 CLAIMS (Prioritized by Novelty × Impact)

CLAIM A: ESI Derivation from Publish/Subscribe Object Identifiers

Prior Art Risk: LOW — strongest candidate

Independent Claim: A method for forward error correction in a publish/subscribe media transport system, comprising:

- a. receiving a media object identified by a group identifier G and an object identifier O within a named track of a publish/subscribe session;
- b. computing an interleave depth D in media units from a time-based interleave parameter expressed in milliseconds and a media frame duration, wherein said millisecond unit decouples the interleave depth from media frame rate;
- c. deriving a Source Block Number $SBN = \text{floor}(G / D)$;
- d. deriving an Encoding Symbol ID $ESI = (G - SBN * D) * \text{ceil}(K / D) + O$, where K is the number of source symbols per FEC block;
- e. using said SBN and ESI to identify the symbol's position within a FEC block for encoding or decoding per RFC 6330;
- f. wherein a flat Source Symbol ID $SS_ID = SBN * K + ESI$ is independently derivable from said SBN and ESI, and said SS_ID is identical to the Source FEC Payload ID carried in MMTP multicast packets per ISO 23008-1 Section C.5.2, enabling a receiver to combine symbols received via the publish/subscribe transport with symbols received via multicast into the same FEC block.

Dependent Claims:

A1. The method of Claim A wherein the interleave depth parameter is specified in milliseconds in a catalog document, and the receiver converts to media units using detected frame duration, making the FEC block mapping independent of video frame rate, audio sample rate, or other media-specific timing.

A2. The method of Claim A wherein a relay node performs said derivation to bridge between a multicast transport carrying SS_ID and a publish/subscribe transport carrying (G, O), translating FEC coordinates between transport domains without modifying the underlying FEC block structure.

A3. The method of Claim A wherein the same derivation is applied independently at multiple receivers and relay nodes, producing identical (SBN, ESI) coordinates for the same media packet regardless of which transport path delivered it.

A4. The method of Claim A wherein the Source Block Number SBN is derived from a shared wall-clock reference as $SBN = \text{floor}((\text{ntp_time} - \text{epoch}) / \text{block_duration})$, where $\text{block_duration} = K * \text{GOP_duration} * D$, and epoch is a session start time signaled in the catalog or AL-FEC signaling message, enabling multiple encoders (hot standby primary/backup or simulcast bitrate tiers) to independently produce identical FEC block boundaries without direct inter-encoder synchronization, such that a receiver combining symbols from any encoder into the same FEC block achieves correct recovery.

A5. The method of Claim A4 wherein the Source FEC Payload ID (SS_ID) carried in MMTP source packets is decoupled from the MMTP packet_sequence_number (PSN), such that an encoder failover that restarts PSN does not disrupt the FEC block namespace, and receivers identify block membership exclusively from SS_ID regardless of PSN discontinuities.

A6. The method of Claim A wherein the Source FEC Payload ID (SS_ID) is carried redundantly in a broadcast hint track sample (source_fec_payload_id field per ISO 23008-1) that falls within the ATSC 3.0 Signed Application (A3SA) signed region per ATSC A/360 Section 5.2.2.5, enabling receivers to verify the integrity of FEC block assignment using broadcast digital signatures without relying on the unsigned trailing 4-byte Source FEC Payload ID appended to asset packets.

Prior Art Analysis: - RFC 6330 §4.4: defines SBN/ESI as FEC coordinates but does NOT define how they map to any transport-layer identifier. Mapping left to the Content Delivery Protocol. - RFC 5775/6726 (ALC/FLUTE): TOI-based mapping for file objects. No formula involving interleave depth or real-time media group/object IDs. - ISO 23008-1 Annex C: defines SS_ID as flat counter with $SBN = \text{floor}(SS_ID / K)$, $ESI = SS_ID \% K$. Does NOT bridge to any pub/sub transport. - US9270299B2 (Qualcomm): flexible source block mapping. Dynamic assignment, not deterministic formula from transport IDs. - **Gap:** No prior art maps pub/sub (group_id, object_id) to FEC (SBN, ESI) with time-based interleave. The bridging formula that produces identical coordinates across MoQ QUIC and MMTP multicast is novel.

Implementation Evidence: - Draft: §8.3 (Source Symbol ESI Derivation) - Code: libmmt/mmt-fec/src/raptorq.rs line 101 `global_ss_id = block_id * K + symbol_id` - Tests: 12 RaptorQ tests pass including recovery with missing source

CLAIM B: Multi-Path FEC Symbol Combining Across Heterogeneous Transports

Prior Art Risk: MEDIUM — novel combination of known techniques

Independent Claim: A method for recovering media data in a streaming system, comprising:

- a. receiving one or more source symbols via a first transport protocol selected from: reliable QUIC streams, unreliable QUIC datagrams, or multicast UDP;
- b. receiving one or more repair symbols via a second transport protocol different from said first transport protocol;
- c. for each received symbol, deriving FEC coordinates (SBN, ESI) that are identical regardless of which transport protocol delivered the symbol, using a deterministic derivation from the symbol's transport-layer identifier;
- d. deduplicating symbols received via multiple transport paths using said (SBN, ESI) coordinates as a uniqueness key;
- e. combining deduplicated source and repair symbols from said heterogeneous transport protocols into a single FEC block;
- f. performing FEC decoding on said combined block to recover lost source symbols when the total number of received symbols (source + repair) meets or exceeds K.

Dependent Claims:

B1. The method of Claim B wherein source symbols are received via multicast UDP and repair symbols are received via reliable QUIC, enabling FEC recovery where the unreliable multicast path provides bulk source data and the reliable unicast path provides targeted repair.

B2. The method of Claim B wherein a receiver maintains a single FEC decoder instance per source block that accepts symbols from any transport path, and said decoder produces recovered data identically regardless of which paths contributed symbols.

B3. The method of Claim B wherein a receiver dynamically switches its primary source from one transport protocol to another while maintaining FEC block continuity, such that symbols received before and after the switch contribute to the same FEC block.

Prior Art Analysis: - US8787153B2 (Cisco): FEC with path diversity — sends source on one path, repair on another. CLOSEST prior art. But addresses spatial diversity between two paths of the SAME transport type (two IP routes), NOT heterogeneous transports (UDP multicast vs

QUIC). - 3GPP TS 26.346 (eMBMS): unicast HTTP used to fetch missing symbols after multicast. Retry/request mechanism, NOT real-time combining. - ATSC 3.0 A/331: broadband HTTP unicast repair as fallback. Same retry pattern as 3GPP. - draft-navarre-quic-flexicast: combines unicast+multicast in QUIC. Both paths use QUIC — not heterogeneous transport. - **Gap:** Real-time symbol combining where one path is reliable QUIC and the other is unreliable multicast UDP, unified by shared FEC coordinates. No prior art addresses this specific combination.

Implementation Evidence: - Draft: §8.3, §11.1 - Code:

`libmmt/packages/container/src/media-router.ts` (source arbitration), `fec-manager.ts` (block combining)

CLAIM C: Hybrid Multicast-Unicast Source Arbitration with FEC-Aware Switching

Prior Art Risk: MEDIUM — novel FEC-aware aspects over known switching

Independent Claim: A method for media reception in a hybrid delivery system, comprising:

- a. subscribing to a media track via a unicast publish/subscribe transport;
- b. simultaneously initiating a multicast group join for the same media content;
- c. receiving and rendering media from the unicast transport during a multicast join period, providing zero-latency startup;
- d. upon receiving a video keyframe (random access point) via the multicast transport, activating multicast as the primary media source and suppressing duplicate unicast frames;
- e. during a transition period between unicast and multicast primary sources, injecting FEC repair symbols received from the unicast transport into FEC blocks whose source symbols were received from the multicast transport, enabling cross-transport FEC recovery;
- f. applying hysteresis to the transport switching decision, wherein the receiver switches away from multicast only after detecting sustained loss exceeding a threshold, and switches back to multicast only after stable reception for a configurable period;

g. upon detecting sustained multicast failure, failing back to unicast by re-subscribing with a LatestGroup delivery preference to resynchronize to the current media position.

Dependent Claims:

C1. The method of Claim C wherein audio multicast activation is gated on a separate timer independent of video, allowing audio to activate via multicast even if video multicast has not yet delivered a keyframe.

C2. The method of Claim C wherein said FEC repair symbol injection (step e) uses the same FEC block coordinates derived from the ESI derivation formula of Claim A, ensuring symbols from both transport paths are correctly assigned to the same FEC block.

C3. The method of Claim C wherein a FEC block lifecycle state machine tracks the completeness of each block across both transport paths, and said state machine determines when to attempt FEC recovery based on combined symbol counts from both transports.

Prior Art Analysis: - US20060200576A1 (Cisco): simultaneous unicast+multicast until duplicate frames buffered, then switch. Does NOT address FEC-aware switching or keyframe gating. - US7788393B2 (Cisco): increases unicast rate to buffer ahead, then switch to multicast. Buffer management, NOT FEC awareness. - EP2013987A1: staggercasting with FEC for handover. Server-side technique (shifted streams), NOT receiver-side arbitration. - 3GPP eMBMS: network-initiated handover (eNodeB decides), NOT receiver-side FEC-aware arbitration. - **Gap:** FEC-aware keyframe gating, cross-transport symbol injection during transition, receiver-side FEC block completeness-based arbitration. These specific FEC mechanisms are not in any found prior art.

Implementation Evidence: - Draft: draft-ramadan-moq-multicast-00 §4.1 - Code: libmmt/packages/container/src/media-router.ts

CLAIM D: FEC Block Lifecycle State Machine with Codec-Aware Timeout

Prior Art Risk: LOW — genuinely novel

Independent Claim: A method for managing forward error correction decoding in a media receiver, comprising:

a. maintaining per-block state with three mutually exclusive outcomes:

- clean: all K source symbols received without loss;
 - recovered: fewer than K source symbols received but FEC decoding using repair symbols successfully recovered the block;
 - failed: a timeout expired with insufficient symbols for recovery;
- b. starting a block timeout upon receipt of the first source symbol for said block, not upon block creation or first repair receipt;
- c. computing said timeout dynamically as a function of:
 - the interleave depth (number of media frames per FEC block),
 - the detected media frame rate, and
 - a configurable safety multiplier;
- d. marking a block as clean when K source symbols are received before the timeout, bypassing FEC decoding entirely for lossless blocks;
- e. attempting FEC decoding when repair symbols arrive for a block that has fewer than K source symbols;
- f. marking a block as failed when the timeout expires with fewer than K total symbols (source + repair);
- g. maintaining a completed-blocks set that prevents late-arriving symbols from restarting timeouts or triggering redundant decoding for blocks that have already reached a terminal state;
- h. maintaining per-track FEC decoder instances keyed by media track identifier, each with independent K, timeout, and block state, enabling different FEC parameters per media type within the same session.

Dependent Claims:

D1. The method of Claim D further comprising refining the timeout computation when the frame rate is first detected from decoder output, and propagating said refined timeout to all active decoder instances.

D2. The method of Claim D wherein the three-state accounting (blocksClean, blocksRecovered, blocksFailed) is exposed as real-time statistics for adaptive bitrate selection or quality monitoring

dashboards.

D3. The method of Claim D wherein a bounded-size completed-blocks set is maintained using insertion-order eviction, keeping the most recent N blocks and evicting the oldest to prevent unbounded memory growth.

Prior Art Analysis: - RFC 6363 (FECFRAME): defines block FEC code framework. Receiver groups packets by FEC Payload ID into source blocks, decodes. Does NOT define timeout behavior, state machines, or codec-aware timing. Receiver behavior explicitly left to implementation. - RFC 6330 (RaptorQ): defines encoding/decoding algorithms. WS parameter limits decoder memory. Does NOT define block timeout, lifecycle states, or receiver timing. - SMPTE 2022-1 (Pro-MPEG FEC): row/column FEC matrix. Defines matrix structure but NOT timeout management or lifecycle state machines. - RFC 8680 (FECFRAME sliding window): mentions source symbol removal after delays. Closer to timeout concept but for sliding window, not block lifecycle. - **Gap:** No found prior art defines a three-state block lifecycle with codec-aware dynamic timeout, completed-block guard, or per-track decoder routing. FEC standards deliberately leave receiver behavior unspecified.

Implementation Evidence: - Code: `libmmt/packages/container/src/fec-manager.ts` (FECManager class, 319 lines, 3-state accounting, dynamic timeout) - Tests: 4 NoneCodec tests, verified in E2E 7-point validation

CLAIM E: End-to-End Broadcast-to-Unicast Bridge with FEC Preservation

Prior Art Risk: LOW-MEDIUM — system claim, novel pipeline

Independent Claim: A system for bridging broadcast media to unicast streaming, comprising:

- a. a broadcast gateway receiving MMTP packets with application-layer FEC via SSM multicast from a broadcast transmitter conforming to ATSC 3.0 or ARIB STD-B60;
- b. said gateway performing FEC decoding on received MMTP packets to recover source media lost during broadcast transmission;
- c. said gateway converting broadcast signaling from S-TSID XML format to a streaming catalog in JSON format, wherein said conversion maps:
 - o S-TSID Logical Stream elements to catalog track entries;

- S-TSID FECParameters to catalog FEC extension fields, with: fecSchemeID mapped to algorithm string, symbolSize mapped to symbolSize integer, maxSourceBlockLength mapped to sourceSymbols integer, (maxNumberEncSymbols - maxSourceBlockLength) mapped to repairSymbols integer;
 - S-TSID RepairFlow elements to catalog repair track entries;
 - S-TSID RS (Receiving Station) elements to catalog multicast endpoint entries;
- d. said gateway publishing recovered source media as tracks in a publish/subscribe session over QUIC or WebTransport;
- e. one or more subscribers receiving said tracks without requiring MMTP decoding capability, FEC processing, or broadcast receiver hardware;
- f. wherein the reverse conversion from streaming catalog to S-TSID is also performed, enabling MoQ-originated content to be retransmitted via broadcast infrastructure.

Dependent Claims:

E1. The system of Claim E wherein the gateway re-encodes recovered source media from MMTP packaging to CMAF packaging, producing fMP4 segments suitable for HLS or DASH delivery to downstream clients.

E2. The system of Claim E wherein the gateway selectively preserves FEC by re-publishing both source and repair tracks for subscribers that request FEC protection, while publishing source-only tracks for subscribers that do not.

E3. The system of Claim E wherein the gateway is deployed at an ISP edge router with inline AMT relay capability, forming a TreeDN node that receives broadcast via AMT tunnel and re-publishes to local subscribers.

Prior Art Analysis: - US20250274499 (CDN offload ATSC 3.0): hybrid DASH broadcast + CDN unicast. Uses DASH/ROUTE, NOT MMT/MMTP. No FEC recovery or MoQ republishing. No S-TSID conversion. - Samsung EP2849440A1: MMT hybrid delivery. Predates QUIC/MoQ. Does not cover S-TSID↔catalog conversion or relay FEC recovery. - ADTH ATSC 3.0 Gateway: commercial product converting OTA to home network. Does NOT preserve FEC or output MoQ. - Harmonic/MediaKind gateways: transcoding gateways. Involve transcoding, not FEC-preserving bridging. - **Gap:** The complete pipeline (SSM → FEC recovery → S-

TSID→catalog conversion → MoQ republish) is not found in any prior art. The S-TSID↔MoQ catalog bidirectional conversion is entirely novel.

Implementation Evidence: - Draft: draft-ramadan-moq-mmt-00 §11 (conversion), §8 (multicast), Appendix B (example) - Draft: draft-ramadan-moq-fec-00 §12 (ATSC compatibility)

CLAIM F: Catalog-Based FEC Discovery with Hierarchical Defaults

Prior Art Risk: LOW — novel signaling model

Independent Claim: A method for forward error correction parameter signaling in a media streaming system, comprising:

- a. a catalog document in JSON format containing:
 - a top-level FEC configuration object specifying default FEC parameters including algorithm, sourceSymbols, repairSymbols, symbolSize, and interleaveDepth;
 - zero or more per-track FEC configuration objects that override specific fields from the top-level defaults, wherein unspecified fields inherit the top-level values;
- b. said catalog document being fetchable via HTTP prior to establishing a streaming session, enabling subscribers to discover FEC parameters before committing to subscription;
- c. an in-band FEC_CONFIG control message sent from publisher to subscriber after subscription establishment, carrying authoritative FEC parameters;
- d. a precedence order wherein: the in-band FEC_CONFIG message overrides per-track catalog values, which override top-level catalog defaults;
- e. wherein a per-track algorithm field enables different FEC schemes per media type within the same catalog, including a “none” value that specifies no FEC encoding with an optional jitterBufferMs parameter for receiver-side reorder tolerance.

Dependent Claims:

F1. The method of Claim F wherein said “none” algorithm with jitterBufferMs enables ultra-low-latency media tracks to coexist with FEC-protected tracks in the same session, with the receiver applying only a reorder buffer (no FEC decode) for “none” tracks.

F2. The method of Claim F wherein deferred FEC configuration is merged incrementally as track renditions are created in the catalog, preserving fields set by earlier configuration calls and applying pending defaults to newly created renditions.

F3. The method of Claim F wherein for multicast delivery where no back-channel exists for the in-band FEC_CONFIG message, the catalog FEC parameters serve as the sole authoritative source.

Prior Art Analysis: - RFC 6364 (SDP FEC): FEC signaling via SDP attributes. Session-level, not manifest/catalog-based. Requires SIP/RTSP session. No hierarchical defaults. - DASH MPD: has hierarchical structure (Period > AdaptationSet > Representation) with property inheritance. But NO FEC-specific properties defined in DASH. - ATSC A/331 S-TSID: per-flow FEC in XML. Not hierarchical JSON. No “none” algorithm. No HTTP pre-discovery. - MoQ catalog (I-D.ietf-moq-catalogformat): JSON catalog for MoQ. Has ZERO FEC provisions. draft-ramadan-moq-fec is the first to propose FEC extensions. - **Gap:** No prior art combines JSON catalog FEC discovery with hierarchical defaults, HTTP pre-subscription access, in-band override precedence, and per-track algorithm selection including “none” mode.

Implementation Evidence: - Draft: draft-ramadan-moq-fec-00 §4.1, §5 - Code: hang-mmt-fec/rs/hang/src/catalog/root.rs (top-level fec field, pending_video_fec inheritance pattern) - Code: hang-mmt-fec/js/hang/src/catalog/root.ts (RootSchema.fec) - Tests: top_level_fec_inherits_to_renditions, per_rendition_fec_overrides_top_level (both pass)

NEW DEPENDENT CLAIMS (Added 2026-04-14)

A4: NTP-Synchronized Block IDs for Hot Standby

Prior Art Risk: LOW — No prior art combines wall-clock-derived FEC block boundaries with pub/sub media transport.

Prior Art Analysis: - DVB BlueBook A176: Defines MPEG-2 TS redundancy switching, but no FEC block synchronization across encoders. Switching is at TS packet level, not FEC block level. - SMPTE ST 2022-7: Seamless protection switching for SMPTE 2110, but operates at RTP/IP layer with hitless merge, not FEC block alignment. - 3GPP TS 26.346 (MBMS): FEC over FLUTE/ALC with SBN assigned by sender. No receiver-derivable SBN from wall clock. -

Gap: Deriving SBN from NTP time so that independent encoders produce identical block boundaries — enabling receiver-side combining of symbols from any encoder — is novel.

Implementation: Draft §8.5, FFmpeg `-fec_ntp_sync` (planned)

A5: PSN-Decoupled SS_ID for Encoder Failover

Prior Art Risk: LOW — PSN and SS_ID are separate fields in ISO 23008-1, but no prior art explicitly claims the decoupling as enabling seamless FEC recovery across encoder transitions.

A6: A3SA Signed FEC Block Assignment

Prior Art Risk: LOW — Carrying SS_ID in the hint track for A3SA signed-region integrity is a novel combination of ISO 23008-1 hint tracks (existing) with ATSC A/360 signing (existing) for FEC block authentication (new use case).

Implementation: Draft §8.6, §14.5

CLAIMS NOT IN TOP 6 (Demoted)

Claim 3 (Relay-Side FEC Recovery) → DEMOTED

Reason: RFC 6363 §8 explicitly describes “decoding middlebox” that performs FEC recovery before forwarding. The core concept has direct prior art. Fold the MoQ-specific pub/sub subscription model into Claims B and E as dependent claims.

Claim 5 (Unified Packet Format) → DEMOTED

Reason: Samsung EP2849440A1 covers hybrid broadcast/broadband using the same MMT packet format. ISO 23008-1 was literally designed for heterogeneous transport. The QUIC stream vs datagram distinction is narrow. Risk of rejection against Samsung’s portfolio is HIGH.

Claim 9 (AMT/DRIAD in Catalog) → NARROW SCOPE

Reason: Combines two existing protocols (AMT RFC 7450 + DRIAD RFC 8777) with catalog signaling. Valuable but the combination may be viewed as obvious. Consider as dependent claim

under Claim F.

Claim 10 (S-TSID Conversion) → FOLDED INTO CLAIM E

Reason: The conversion algorithm is the core of Claim E (broadcast bridge). Stronger as part of the system claim than standalone.

FILING STRATEGY

Application 1: Core FEC Innovations (File First)

Claim	Title	Risk
A (independent)	ESI Derivation from Pub/Sub IDs	LOW
B (independent)	Multi-Path FEC Combining	MEDIUM
C (independent)	Hybrid Source Arbitration	MEDIUM
D (independent)	FEC Block Lifecycle State Machine	LOW
A1-A3	ESI derivation dependents	—
A4	NTP-Synchronized Block IDs (hot standby)	LOW
A5	PSN-Decoupled SS_ID (encoder failover)	LOW
A6	A3SA Signed FEC Block Assignment	LOW
B1-B3, C1-C3, D1-D3	Other dependent claims	—

Rationale: Claims A and D are the strongest (lowest prior art risk). Claims B and C are enabled by Claim A (the ESI derivation formula is what makes multi-path combining and hybrid switching work). A4-A6 strengthen Claim A with hot standby synchronization and broadcast signing — critical for carrier-grade deployment and ATSC3 compliance. Filing together creates a coherent patent on “FEC for publish/subscribe media transport with multi-path delivery.”

Application 2: System Claims (File Second)

Claim	Title	Risk
E (independent)	Broadcast-to-Unicast Bridge	LOW-MEDIUM
F (independent)	Catalog-Based FEC Discovery	LOW
E1-E3, F1-F3	Dependent claims	—

Rationale: E and F are system/signaling claims that complement the core FEC methods. E covers the full broadcast bridge pipeline; F covers the catalog signaling that enables all other claims. Filing as a continuation-in-part preserves the priority date.

Prior Art to Distinguish in Prosecution

Prior Art	What It Covers	Our Novel Contribution
RFC 6330 (RaptorQ)	FEC algorithm	Transport mapping (Claim A)
RFC 6363 (FECFRAME)	FEC framework, middlebox	Pub/sub model, block lifecycle (Claims B, D)
US8787153 (Cisco)	FEC path diversity	Heterogeneous transport combining (Claim B)
US20060200576 (Cisco)	Unicast+multicast switching	FEC-aware keyframe gating (Claim C)
Samsung EP2849440A1	MMT hybrid delivery	QUIC binding, MoQ integration (Claims A, B)
ISO 23008-1	MMTP packet format	Pub/sub ESI derivation (Claim A)
ATSC A/331	Broadcast FEC signaling	JSON catalog, bidirectional conversion (Claims E, F)
ATSC A/360 (A3SA)	Broadcast signing	Signed FEC block assignment via hint track (Claim A6)
SMPTE ST 2022- 7	RTP hitless switching	NTP-derived FEC block sync, not RTP merge (Claim A4)

Prior Art	What It Covers	Our Novel Contribution
DVB BlueBook A176	TS redundancy switching	FEC-block-level failover, not TS packets (Claim A4)
3GPP TS 26.346 (MBMS)	FLUTE/ALC FEC	Receiver-derivable SBN from wall clock (Claim A4)
RFC 6364	SDP FEC signaling	Hierarchical JSON catalog (Claim F)

APPENDIX B — Intrinsic Fragment Coordinate (IFC): Design Disclosure

Intrinsic Fragment Coordinate (IFC) — design note

Status: design sketch (not a submission draft). Source material for an extensions-draft section and a patent continuation. **Renamed** from “Generic Fragment Descriptor” — *GFD* collides with MMT’s **Generic File Delivery** mode (ISO/IEC 23008-1 §9.3.3).

Companions: `draft-ramadan-moq-mmt-00.md` (§“MFU Fragmentation”), `draft-ramadan-moq-fec-00.md` ((SBN, ESI) symbol coords), `patent/APPLICATION-1-FEC-METHODS.md`, and **ISO/IEC 23008-1:2023** (verified ground truth — see §2).

1. Problem

Reassembly of a *fragmented* application object must survive transport disorder, loss, FEC recovery, multi-path combining, and relay re-sequencing — without coupling the reassembly layer to a specific media format. We want one reassembler that works for MMT timed media, MMT non-timed items, generic files, and future object types, over a MoQ + multicast + FEC transport.

2. ISO/IEC 23008-1 ground truth (verified)

MMT **already** defines intrinsic, content-carried fragment coordinates — three of them, one per object class. IFC is the *unification* of these; it does not invent new descriptors.

MMT object class	Where	Object key	Intrinsic offset/order	Extras
Timed media MFU	§9.3.2.3, Fig 12 (T=1)	MPU_sequence_number + movie_fragment_sequence_number + sample_number	offset (within the referenced sample)	subsample_priority, dependency_counter
Non-timed media MFU	§9.3.2.3, Fig 13 (T=0)	Item_ID	(item-scoped)	subsample_priority, dependency_counter
Generic File Delivery	§9.3.3 / §9.3.3.4	packet_id (Asset) + TOI	start_offset (48-bit, “location of payload in the object”)	CodePoint

Fragmentation framing is also ISO-native: f_i (00 complete / 01 first / 10 mid / 11 last), aggregation_flag, fragment_counter (§9.3.2.3). The MMT draft’s FI=1/2/3 reassembly is a direct restatement of this.

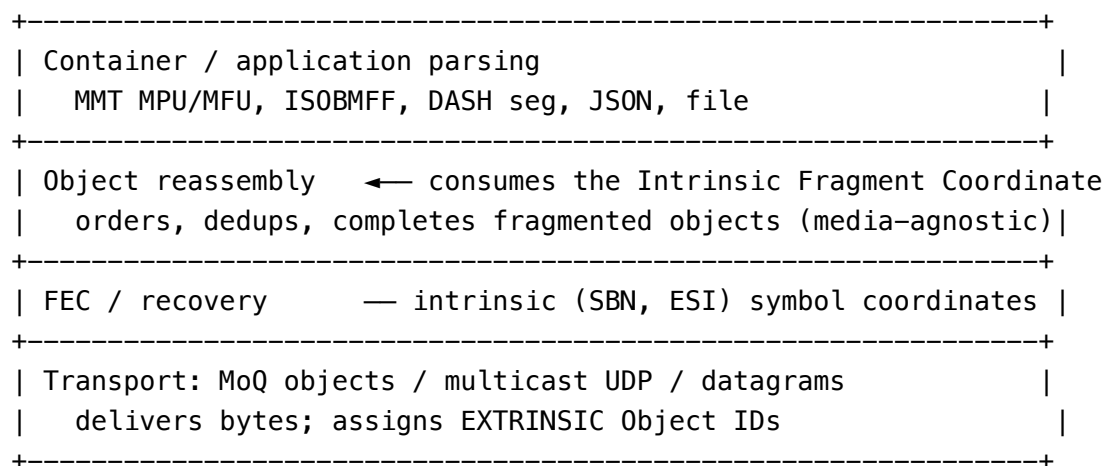
Consequence for libmmt #78: parsing movie_fragment_sequence_number / sample_number / offset is *exactly* §9.3.2.3 — **spec-conformant, not novel**. That’s the right outcome for the implementation; it just means the *descriptors* carry no patent weight (see §9).

3. Principle

Order and dedup fragments by **intrinsic** (content-carried) coordinates — never by **extrinsic** (transport-assigned) delivery order. This mirrors the FEC layer, which already addresses symbols by (SBN, ESI) from SS_ID, not by arrival. IFC applies the same idea at the application-object

boundary. Extrinsic ordering (MoQ Object ID) remains the **mandatory floor** for opaque/unfragmented objects.

4. Layer position



IFC sits **above** FEC (reassembles recovered source bytes) and **below container parsing** (hands up a complete, ordered object). The container layer never sees loss or ordering.

5. The coordinate (abstract; MMT supplies the concrete encodings)

Field	Meaning	MMT source
objectKey	transport-independent identity of the logical object	timed: MPU_seq:MFSN:sample · non-timed: Item_ID · file: packet_id:TOI
fragmentOffset	byte offset within the object	timed: DU offset · file: start_offset
completion	objectTotalLength or last-fragment (f_i=11)	f_i / fragment_counter
rap	object is a random-access point (decode-gate)	MPU first AU = SAP (§6.4); RAP packets carry full headers (§9)

Field	Meaning	MMT source
priority	recovery/decode precedence	subsample_priority, priority_id, transmission_priority
dependency	DUs depending on this one	dependency_counter
contentHash (<i>opt</i>)	path-independent dedup/integrity	— (transport overlay)

Invariants - Order = ascending fragmentOffset (ignores arrival order and Object ID). - Dedup = (objectKey, fragmentOffset) (and/or contentHash). - Completion = contiguous [0, total) coverage. - Bounded memory = discard incomplete objects on a derived deadline.

6. RAP / I-frame handling (per ISO)

- An MPU begins at a **SAP** (§6.4): the first access unit of an MPU is a random-access point. So keyframe boundaries = MPU/Group boundaries.
- RAP is signalled by **full-header packets** (§9): the reassembler's rap bit derives from this, not from a payload bit alone.
- I-frames are **high subsample_priority and dependency roots** (high dependency_counter). IFC surfaces rap + priority + dependency so the receiver can: (a) **gate decode start** on a recovered RAP (patent Emb. 3, state 402a “keyframe gate”); (b) **prioritize FEC recovery** of RAP/high-priority fragments; (c) drop low-priority dependents first under pressure.

7. Precedence model (intrinsic-first, extrinsic-floor)

1. Fragments carry an IFC → reassemble by fragmentOffset, dedup by (objectKey, offset). Robust on FEC / multi-path / relay.
2. No IFC (opaque payload) → fall back to transport **Object ID** order.
3. Whole/unfragmented object → no reassembly; Object ID for delivery + dedup.

This is the shape already in `libmmt mfu-reassembler.ts` (offset-mode when sample identity present, else counter), generalized off MMT.

8. Draft framing (extensions draft, not lean core)

Section: “**Self-Describing Fragment Reassembly.**” - Normative: a receiver MAY use an MMT-native intrinsic coordinate (timed-MFU offset, GFD start_offset, or Item_ID), when present, as the authoritative reassembly order, **taking precedence over the MoQ Object ID**; it MUST fall back to Object-ID order when absent. - Cite ISO/IEC 23008-1 §9.3.2.3 (DU headers) and §9.3.3 (GFD start_offset) as the field source — the draft only specifies the **MoQ mapping + precedence rule**, not new fields. - Updates draft-ramadan-moq-mmt-00 §“MFU Fragmentation”: its “order by Object ID” becomes the *floor*, with intrinsic-offset precedence on top. Lands in the **extensions** draft per the lean-core/low-exposure sequencing.

9. Patent-claim outline (re-scoped above ISO prior art)

ISO already defines the descriptors and offset-reassembly (§2), so claims **must not** read on “use sample/offset/Item_ID/start_offset to reassemble.” The novelty is the **transport reconciliation on a pub/sub + multicast + FEC path** — the up-layer companion to APPLICATION-1’s (SBN, ESI) claims.

- **Independent claim.** A method for reassembling a fragmented application object delivered over a publish/subscribe transport that assigns extrinsic per-object delivery identifiers (e.g., MoQ Object IDs) and over one or more additional paths including multicast with forward error correction, comprising: recovering missing fragments via FEC such that recovered fragments lack a transport-assigned delivery identifier; ordering and deduplicating fragments — including FEC-recovered and multi-path-delivered fragments — by an **intrinsic content-carried coordinate** (object identity + byte offset) **in precedence over** the transport-assigned delivery identifier; signalling completion by contiguous byte coverage; thereby reassembling deterministically irrespective of arrival order, path, or recovery.
- **Dependent claims.**
 - intrinsic coordinate sourced from an MMT timed-MFU DU header (movie_fragment_sequence_number/sample_number/offset);
 - ...from an MMT non-timed Item_ID;
 - ...from an MMT Generic File Delivery start_offset/T0I;
 - ...from a transport object **extension header** for non-MMT payloads (the generalization);
 - dedup of multi-path fragments by content hash;
 - **gating decode start** on a FEC-recovered random-access fragment;
 - **prioritizing FEC recovery** by subsample_priority/dependency_counter;

- bounded-memory discard on a derived deadline;
- composition with (SBN, ESI) symbol addressing of APPLICATION-1.

Positioning: broader and more standards-essential than MMT-only, because the claimed step is the *intrinsic-over-extrinsic reconciliation across FEC/multi-path*, which MMT-over-broadcast never needed (no MoQ Object ID, no unicast/multicast FEC combining) and which ISO therefore does not anticipate. The nearest in-MoQ reference, **draft-ietf-moq-loc-03 §4.5**, establishes a timestamp-primary/ObjectID-derived-fallback (layer-interleave constants, not byte offsets) *precedence pattern* but only for whole, in-order, single-path frames (no fragmentation, FEC, or multi-path); the claims extend that precedence to the fragmented + FEC-recovered + multi-path regime rather than reading on it (see §10.1).

10. Relationship to existing work

- **Conforms to** ISO/IEC 23008-1 §9.3.2 / §9.3.3 (uses its descriptors verbatim).
- **Extends** draft-ramadan-moq-mmt-00 §“MFU Fragmentation” (adds intrinsic precedence over MoQ Object ID; generalizes off MMT).
- **Aligns with** the FEC layer’s (SBN, ESI) intrinsic addressing — IFC is the same idea one layer up.
- **Distinct from** APPLICATION-1 claims (FEC-symbol dedup); IFC is application-*object* reassembly over the FEC/MoQ transport → candidate continuation.

10.1 Relationship to existing MoQ drafts (nearest prior art)

The IFC precedence model (§7) is **not without precedent in MoQ** — and saying so sharpens, not weakens, the positioning. The nearest prior art is **draft-ietf-moq-loc-03 §4.5**, whose worked example (non-normative, dyadic-framerate layered tracks) orders decode as **timestamp primary** with **ObjectID * multiplier + offset** as the fallback ordering key. That is the *same precedence shape* as IFC: an intrinsic-ish ordering key the receiver uses instead of raw arrival order. The distinction is **regime, not shape**:

- **draft-ietf-moq-loc-03 §4.5** — the precedence pattern, but only for **whole frames**. LOC objects are never fragmented; there is no FEC recovery, no multi-path combining, and no relay re-sequencing in its scope. So LOC establishes the timestamp/ObjectID precedence for in-order, single-path, whole-frame delivery. **IFC generalizes that precedence to the fragmented + FEC-recovered + multi-path regime**: (a) sub-frame MFUs/DUs rather

than whole frames; (b) FEC-recovered fragments, which carry no transport sequence yet must slot into the right intrinsic position; (c) multi-path / relay-re-sequenced delivery, where the extrinsic Object ID is unreliable. The intrinsic byte-offset (or sample/DU) coordinate is what survives all three — exactly where LOC’s ObjectID-derived fallback cannot reach.

- **draft-ietf-moq-msf §6.1** — *adjacent but not reassembly*. MSF §6.1’s Prior Group ID Gap Extension marks **intentional numbering discontinuities** (publisher restart — nothing was transmitted, so there is nothing to recover); IFC addresses **loss-induced** gaps and closes them (FEC + offset reassembly) rather than only marking them.
- **MoQ/QUIC FEC** (draft-michel-quic-fec, draft-zheng-quic-fec-extension) and **RFC 6363 FECFRAME** operate at the **QUIC/symbol layer only**. None define object reassembly by an intrinsic content coordinate; IFC sits one layer above them (§4), consuming recovered source bytes.

In short: IFC = the LOC §4.5 timestamp/ObjectID precedence pattern **extended** to fragmented, recovered, and multi-path objects — not a wholly novel idea with no precedent.

APPENDIX C — IFC Draft Claim Language

Intrinsic Fragment Coordinate reassembly — draft claims (for review)

Status: draft claim language for review — *not* a filed application. Companion / continuation to patent/APPLICATION-1-FEC-METHODS.md.

Novelty scoping (read first). ISO/IEC 23008-1:2023 already defines the *descriptors* — the timed-MFU DU header (§9.3.2.3: `movie_fragment_sequence_number`, `sample_number`, `offset`), the non-timed `Item_ID`, and the Generic File Delivery `start_offset` (§9.3.3) — and offset-based reassembly within each. These are prior art and the claims below do **not** read on them in isolation. The claimed advance is the **reconciliation**: using an intrinsic, content-carried coordinate, *in precedence over a transport-assigned object identifier*, to order and de-duplicate fragments **including FEC-recovered and multi-path-combined fragments**, over a publish/subscribe plus multicast transport. Broadcast MMT has no MoQ Object IDs and no

unicast/multicast FEC combining, so ISO does not anticipate this step. The claim set is the application-object-layer companion to APPLICATION-1's FEC-symbol-layer (SBN, ESI) claims.

Distinction over the nearest MoQ prior art (draft-ietf-moq-loc-03 §4.5). The closest reference within MoQ is draft-ietf-moq-loc-03 §4.5, whose worked example (non-normative, dyadic-framerate layered tracks) orders decode as timestamp-primary with `ObjectID * multiplier + offset` as the fallback ordering key — i.e., the same *precedence shape* the claims rely on (an intrinsic-ish ordering key used in place of raw arrival order). The claims are framed as **extending**, not contradicting, that pattern. LOC §4.5 applies only to **whole, in-order, single-path frames**: LOC objects are never fragmented, and LOC defines no FEC recovery, no multi-path combining, and no relay re-sequencing. The claimed method operates precisely in the regime LOC does not reach — **fragmented** (sub-frame) objects, **FEC-recovered** fragments that carry no transport-assigned identifier (step (b)/(c)), and **multi-path-combined** fragments where the `ObjectID`-derived key is path-relative and unreliable (Claim 6). In all three the intrinsic byte offset is the only key that survives, so LOC's whole-frame `ObjectID`-offset fallback does not anticipate step (c). draft-ietf-moq-msf §6.1 is adjacent only — it marks intentional publisher-restart numbering discontinuities (loss-independent, nothing to recover) but performs no reassembly. None of draft-michel-quic-fec, draft-zheng-quic-fec-extension, or RFC 6363 FECFRAME define object reassembly by an intrinsic content coordinate; they protect the QUIC/symbol layer beneath this method.

Independent Claim 1 (method)

A method for reassembling a fragmented application object delivered to a receiver over (i) a publish/subscribe transport that assigns to delivered objects an extrinsic, transport-assigned delivery identifier, and (ii) at least one further delivery path comprising multicast delivery protected by forward error correction (FEC), the method comprising:

- a. receiving a plurality of fragments of the application object, each fragment carrying an intrinsic descriptor comprising an object identity and a byte offset of the fragment within the application object, the intrinsic descriptor being independent of the transport-assigned delivery identifier;
- b. recovering at least one missing fragment of the application object by FEC decoding, the recovered fragment lacking a transport-assigned delivery identifier;

- c. ordering the fragments, including the FEC-recovered fragment, by the byte offset of the intrinsic descriptor, in precedence over the transport-assigned delivery identifier;
- d. de-duplicating the fragments by the intrinsic descriptor, such that a fragment whose object identity and byte offset have already been incorporated is discarded irrespective of the delivery path on which it arrived;
- e. determining completion of the application object by contiguous byte coverage; and
- f. outputting the reassembled application object to a container or application parser;

whereby the application object is reassembled deterministically irrespective of fragment arrival order, delivery path, or FEC recovery.

Dependent Claim 2

The method of Claim 1, wherein the intrinsic descriptor is carried in a media data unit header of an MPEG Media Transport (MMT) timed-media media fragment unit and comprises a movie fragment sequence number, a sample number, and an offset within the referenced sample.

Dependent Claim 3

The method of Claim 1, wherein the intrinsic descriptor is carried in a media data unit header of an MMT non-timed media fragment unit and comprises an item identifier.

Dependent Claim 4

The method of Claim 1, wherein the intrinsic descriptor is carried in a generic file delivery payload header and comprises a transport object identifier and a start offset.

Dependent Claim 5

The method of Claim 1, wherein the application object is not an MMT object and the intrinsic descriptor is carried in an extension header of a transport object of the publish/subscribe transport.

Dependent Claim 6

The method of Claim 1, wherein the further delivery path and the publish/subscribe transport deliver the same application object, and de-duplicating in step (d) comprises discarding a fragment combined from a second delivery path whose intrinsic descriptor matches that of a fragment already incorporated from a first delivery path.

Dependent Claim 7

The method of Claim 6, wherein de-duplicating further comprises comparing a content hash carried with, or computed over, each fragment.

Dependent Claim 8

The method of Claim 1, further comprising gating a start of media decoding on the completion — including completion by FEC recovery — of an application object that is a random access point.

Dependent Claim 9

The method of Claim 1, further comprising prioritising FEC recovery or buffer retention of fragments according to a priority value or a dependency count carried in, or associated with, the intrinsic descriptor.

Dependent Claim 10

The method of Claim 1, further comprising discarding an incompletely-covered application object upon expiry of a deadline derived from a FEC interleave depth or a frame duration, thereby bounding reassembly memory.

Dependent Claim 11

The method of Claim 1, wherein the FEC decoding of step (b) operates on source symbols addressed by intrinsic FEC coordinates derived as in the method of APPLICATION-1 Claim 1,

such that both the FEC symbol layer and the application-object reassembly layer order and deduplicate by intrinsic, transport-independent coordinates.

Independent Claim 12 (apparatus)

A receiver comprising one or more processors and a memory storing instructions that, when executed by the one or more processors, cause the receiver to perform the method of any of Claims 1 to 11.

Independent Claim 13 (medium)

A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any of Claims 1 to 11.

Notes for the drafter

- Claim 1 step (c) “in precedence over the transport-assigned delivery identifier” is the crux distinguishing over ISO/IEC 23008-1 (which has no competing transport identifier) and over plain MoQ (which orders by Object ID). Step (c) alone does not distinguish over the nearest MoQ reference, **draft-ietf-moq-loc-03 §4.5**: LOC §4.5 already pairs a primary ordering key with an ObjectID-derived fallback, but only for whole, in-order, single-path frames. The distinguishing limitations are therefore the *fragmented* + *FEC-recovered* + *multi-path* context — step (b) (recovered fragment lacking a delivery identifier) and Claim 6 (multi-path combining) — which place the method outside LOC’s regime. Keep step (c) in the independent claim and lean on (b)/Claim 6 when arguing non-anticipation over LOC. The prose “Distinction over the nearest MoQ prior art” block above is intended to carry this argument without rewording the claim limitations.
- Claims 2–5 enumerate the descriptor encodings (MMT timed / non-timed / GFD / generic extension header). 2–4 are ISO-sourced encodings; 5 is the generalization that extends reach to JSON/file/arbitrary objects — likely the most commercially important dependent claim.
- Claim 11 is the explicit tie to APPLICATION-1; consider whether to file as a continuation-in-part of APPLICATION-1 or a standalone application that references it.

- Consider a method claim variant where step (b) is omitted (no FEC) but multi-path combining remains, to cover the unicast-multicast dedup case without FEC.
-

APPENDIX D — Self-Describing Fragment Reassembly (Specification Section Draft)

Self-Describing Fragment Reassembly

Motivation

{{mfu-fragmentation}} reassembles the fragments of a Media Fragment Unit (MFU) in reconstructed Object ID order ({{object-identifier-reconstruction}}). The Object ID is assigned by the transport. It is authoritative on a reliable, in-order delivery path, but it is not authoritative on the paths this mapping exists to serve:

- A fragment recovered by FEC ({{fec-integration}}) is reconstructed below the MoQ object layer and carries no transport-assigned Object ID; its position in the object must be re-derived.
- When the same content is combined from more than one path — for example a reliable unicast subscription and a multicast group — Object IDs are path-relative and cannot serve as a common ordering or de-duplication key.
- A publisher MAY emit a constant zero Object ID delta and rely on a relay to re-sequence on egress ({{object-identifier-reconstruction}}); end to end the Object ID is then not a stable content identifier.

MMT data units already carry an intrinsic, content-carried coordinate that is invariant to all of the above. A receiver that treats that coordinate as the reassembly authority — falling back to Object ID order only when it is absent — reassembles deterministically irrespective of arrival order, delivery path, or FEC recovery.

Intrinsic Fragment Coordinate

For each class of MMT object, ISO/IEC 23008-1:2023 defines an intrinsic coordinate comprising a transport-independent object identity and a byte offset. No addition to the MoQ object encoding is required; the fields are already present in the data unit (DU) header or the payload header.

Object class	Object identity	Byte offset	Reference
Timed-media MFU	MPU_sequence_number + movie_fragment_sequence_number + sample_number	offset (in the referenced sample)	23008-1 9.3.2.3
Non-timed media MFU	Item_ID	item-scoped	23008-1 9.3.2.3
Generic File Delivery	packet_id + Transport Object Identifier (TOI)	start_of fset	23008-1 9.3.3

An object whose fragments carry such a coordinate is said to be **self-describing**.

Receiver Behaviour

A receiver reassembling a fragmented object:

1. If every fragment of the object carries an intrinsic byte offset, the receiver **SHOULD** order the fragments by ascending byte offset, and **MUST** treat that order as authoritative in precedence over reconstructed Object ID order.
2. If the intrinsic coordinate is absent, the receiver **MUST** fall back to reconstructed Object ID order (`{{object-identifier-reconstruction}}`).
3. The receiver **MUST** de-duplicate fragments by the pair (object identity, byte offset). A fragment whose coordinate has already been incorporated **MUST** be discarded, including a duplicate that arrives on a different path or that is produced by FEC recovery.
4. The receiver **MUST** determine completion by contiguous byte coverage of the object and **MUST NOT** emit a partially-covered object to the container or application parser.
5. The receiver **MUST** bound the memory used for in-flight reassembly and **MUST** discard an object whose coverage does not complete within a deadline derived from the FEC

interleave depth or the frame duration, rather than buffer without limit (consistent with `{{mfu-fragmentation}}`).

Random Access and Priority

An MPU begins at a Stream Access Point: the first access unit of an MPU is a random access point (RAP) (ISO/IEC 23008-1:2023, 6.4). Accordingly a receiver:

- SHOULD gate the start of media decoding on the availability of a complete RAP object, including a RAP object completed by FEC recovery; and
- SHOULD prioritise FEC recovery and retention of fragments that carry a higher `subsample_priority` or a higher `dependency_counter` (ISO/IEC 23008-1:2023, 9.3.2.3), since these are decode-order roots upon which other access units depend.

Relationship to Object Identifier Reassembly

This section refines `{{mfu-fragmentation}}`. Reconstructed Object ID order remains the mandatory reassembly order for objects that have no intrinsic coordinate — for example opaque or unfragmented objects — and is the universal fallback. Where an intrinsic coordinate is present it takes precedence, because it is invariant to the transport conditions (loss, multi-path combining, FEC recovery, and relay re-sequencing) under which Object ID order is not reliable.

The mechanism is not specific to MMT: the same receiver procedure applies to any fragmented application object whose fragments carry an object identity and a byte offset, whether sourced from an MMT DU header, an MMT Generic File Delivery payload header, or a transport object extension header defined for a non-MMT payload.

Relationship to draft-ietf-moq-loc

The precedence rule defined here — an intrinsic, content-carried key taken in precedence over the transport Object ID — is not new to MoQ. `{{?I-D.ietf-moq-loc}}` the worked example in Section 4.5 of draft-ietf-moq-loc-03 (non-normative, dyadic-framerate layered tracks) already orders decode as timestamp-primary, with the quantity `ObjectID * multiplier + offset` as the fallback ordering key. That is the same precedence shape this section uses: a content-carried ordering key applied in place of raw arrival order.

This section extends that pattern to a regime draft-ietf-moq-loc does not address. A draft-ietf-moq-loc object is a whole frame; it is never fragmented, and that draft defines no FEC recovery, no multi-path combining, and no relay re-sequencing. Its ObjectID-derived fallback is therefore sufficient for whole, in-order, single-path delivery. The procedure in this section instead orders and de-duplicates the **fragments** of an object — including fragments recovered by FEC, which carry no Object ID ({{fec-integration}}), and fragments combined from more than one path, where the Object ID is path-relative — by the intrinsic byte offset. The intrinsic byte offset is invariant to all three conditions, where the Object-ID-derived key of draft-ietf-moq-loc-03 Section 4.5 is not. The two are consistent: where an object is whole and single-path, the receiver behaviour here reduces to Object-ID order, the universal fallback above.

{{?I-D.ietf-moq-msf}} Section 6.1 is related but distinct: its Prior Group ID Gap Extension marks intentional publisher-side numbering discontinuities (e.g. publisher restart), where nothing was transmitted and no recovery applies. The receiver behaviour here addresses loss-induced gaps instead, closing them by FEC and offset reassembly rather than only reporting them. FEC mechanisms defined at the QUIC or symbol layer (for example {{?I-D.michel-quic-fec}} or FECFRAME {{?RFC6363}}) protect the bytes consumed by {{fec-integration}}; they do not reassemble application objects by an intrinsic content coordinate, which is the function this section adds above them.